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Time domain reflectometry: a seminal technique for measuring mass and energy in soil

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Abstract

Soil water exerts a strong influence on the transfer and storage of solutes, heat, air, and even water itself, within the soil profile. Soil water also dominates the mass and energy balance of the soil–atmosphere interface. Over the last decade or so, the development and continuing refinement of the time-domain reflectometry (TDR) technique for in situ, nondestructive measurement of water, ionic solutes and air has revolutionized the study and management of the transfer and storage of mass and energy within the soil profile. TDR-measured water content has been applied successfully to water balance studies ranging from the km scales of small watersheds to the mm scale of the root–soil interface. TDR-measured ionic solute status, which applies to the same sample volume as the water content measurement, has been used successfully on soil column, field plot and whole field scales for in situ determination of solute transport parameters, such as pore water velocity and dispersivity. TDR-measurement of air-filled porosity in space and time has given new insights into the mechanisms controlling aeration and gaseous exchange in the crop root zone. The combined water content – solute mass measurement capability of TDR has made this technique a very powerful tool for characterizing solute leaching characteristics, as well as for evaluating solute transport theories and solute transport models. The portability of TDR instrumentation coupled with the simplicity and flexibility of TDR soil probes has allowed the separation of water and solute content measurement error from soil variability, resulting in the capability for determining the mechanisms behind the spatial and temporal variability in field-based soil water content distributions and solute leaching patterns. The usefulness and power of the TDR technique for characterizing mass and energy in soil is increasing rapidly through continuing improvements in operating range, probe design, multiplexing and automated data collection. © 1998 Elsevier Science B.V. All rights reserved.

Keywords: TDR; Mass; Energy; Water content; Solute leaching; Soil aeration; Root zone

1. Introduction

Although most biological mass and energy in the crop root zone is ultimately derived from incident solar radiation, it is the amount, distribution, availability and interactions of water, solutes and air in the

root zone that exert the major influence over how this mass and energy is stored and transferred. For example, when soil water content is high but not excessive, plants can grow well because transpiration and photosynthesis are not restricted, and greater masses of nutrient are available to plants through dissolution. Excessive water contents, on the other hand, can impede crop growth by restricting the exchange of

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soil air with the atmosphere and by keeping the soil temperature low. Soil water moves faster under high water contents (higher hydraulic conductivity), which results in more rapid drainage and perhaps more extensive leaching of dissolved crop nutrients. Lack of nutrients, in turn, impedes crop growth through reduced fertility. Low water contents can restrict water and nutrient availability to crops, as well as allow excessive temperatures in the root zone. High or low soil temperatures, as well as low soil air contents, can impede crop growth regardless of the soil water content and nutrient availability. The interactions among soil water, solute and air are clearly very complex, and their impact on the soil-based mass and energy balances associated with crop production is of crucial importance.

Despite the importance of water, ionic solutes and air in the mass and energy balances of the soil profile, it has been only in the last 20 years that rapid, *in situ*, nondestructive measurement of soil water content, ionic solute concentration and soil air content has become possible in the form of time domain reflectometry (TDR) (Dalton, 1992; Topp et al., 1994; Zegelin et al., 1992). This radar-based technology has revolutionized our ability to characterize the storage and movement of water, solute and air in the soil profile. With TDR it is now practical to monitor simultaneously the soil water, ionic solutes and air (indirectly) in both space and time with high accuracy and relatively low equipment and labour costs. This capability is in turn providing better evaluations of the impacts of agricultural practices on the soil–plant–atmosphere continuum, as well as greater understanding of the mechanisms controlling the mass and energy balances within the soil profile and at the soil–atmosphere interface. This paper gives a brief overview of the operating principles of TDR, as well as a review of how TDR is used to measure water, solute and air in the soil profile. Selected studies will also be reviewed which demonstrate how TDR has improved our understanding of the roles water, solutes and air play in the mass and energy balance of the soil root zone.

2. Measurements with TDR

The high dielectric constant or relative permittivity of water (about 80) compared to that of the other soil

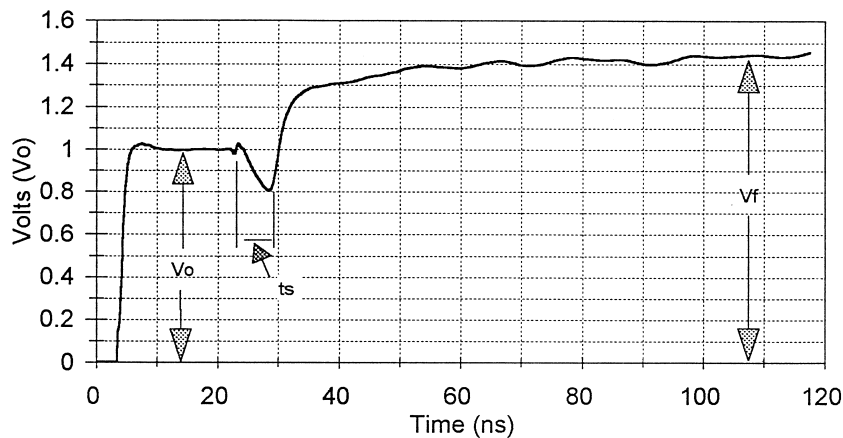
components (1 for air and 2–5 for soil solids) makes determination of the relative permittivity an attractive way to measure water content. The TDR approach, which is a radar technique applied within the soil, is used to determine the soil's bulk relative permittivity. In TDR, a fast rise step voltage pulse is propagated along a transmission line in the soil. The voltage pulse propagates as an electromagnetic (EM) wave, which travels in the soil and is guided by the conductors of a probe, that may have a variety of configurations. The properties of the soil that govern the propagation of EM waves are expressed in terms of velocity, v , and attenuation, α , which can be written as (Topp et al., 1980, 1988),

$$v = \frac{c}{\sqrt{\epsilon_{ra}}} = \frac{2L}{t} \text{ from which } \sqrt{\epsilon_{ra}} = \frac{ct}{2L} \quad (1)$$

$$\alpha = \frac{60\pi(\omega\epsilon_0\epsilon_r'' + \sigma_0)}{\sqrt{\epsilon_{ra}}} \quad (2)$$

where c is the velocity of an EM wave in free space (vacuum), ϵ_{ra} is the apparent relative permittivity measured by TDR, L is the length of the transmission line (probe) in the soil, t is travel time of the EM wave along the probe, ω is the frequency of the propagating wave, ϵ_r'' is the imaginary component of the relative permittivity, σ_0 is the zero frequency (dc) electrical conductivity of the bulk soil, and ϵ_0 is the permittivity of free space. By applying Eqs. (1) and (2) sequentially both water content and bulk electrical conductivity are determined by analyses of TDR traces (Fig. 1) (Topp et al., 1980, 1988). For TDR measurements in soil, ϵ_r'' can be usually neglected without incurring large error (Dalton, 1992; Topp et al., 1988).

An empirical relationship between relative permittivity, ϵ_{ra} , and volumetric water content, θ was initially used for conversion of TDR data to θ (Topp et al., 1980). A variety of researchers have confirmed that the "Topp equation" is quite broadly applicable (Zegelin et al., 1992; Whalley, 1993). Later improvements and refinements have made use of dielectric mixing formulae that require more prior knowledge of the soil properties, such as density, texture and/or organic matter content (Roth et al., 1992; Malicki et al., 1996). From these analyses and related experimental work (Whalley, 1993; White et al., 1994; Malicki et al., 1996), it has been shown that $\sqrt{\epsilon_{ra}}$ is essentially



For 0.2 m length probe - $t_{air} = 1.33$ ns, $Z_0 = 228$ ohms

$$\theta = 0.115 \frac{t_s}{t_{air}} - 0.176 = 0.34$$

$$\sigma_0 = \frac{\epsilon_0 c}{L} \frac{Z_0}{Z_i} \left(\frac{2V_0}{V_f} - 1 \right) = 0.0234 \text{ S m}^{-1}$$

Fig. 1. A representative TDR trace showing how the travel time, t_s , is determined for water content, θ , calculations. Also V_0 and V_f are given for the calculation of bulk conductivity, σ_0 . The other symbols are defined in the text.

a linear function of the water content over the usual water content range. From the assumption that the dielectric properties of the soil phases (solids, water and air) are additive, a further simplification of the calibration approach has been achieved where θ is related linearly to the travel time of the TDR signal (Ledieu et al., 1986; Herkelrath et al., 1991; Hook and Livingston, 1996). The ratio of the TDR travel time in soil to that measured in air, t/t_{air} , is equivalent to $\sqrt{\epsilon_{ra}}$ which means that calibrations for water content measurements can be developed from as few as two measurements.

The use of TDR for electrical conductivity measurements developed rapidly after its introduction (Dalton et al., 1984; Dalton, 1992; Topp et al., 1988). The approaches, now in use to convert TDR data to electrical conductivity, σ_0 are equivalent to that of Giese and Tiemann (1975); (Topp et al., 1988; Zegelin et al., 1989; Heimovaara, 1992; Baker and Spaans, 1993). This approach depends on a knowledge of the characteristic impedance of the TDR probe, Z_0 , output impedance of the TDR instrument, Z_i and impedance of voltage measurements from the TDR waveform (Fig. 1). This analysis is often applied for rapid electrical conductivity measurement during

solute transport experiments and for the determination of mass flux of solute, as is discussed below.

3. Mass balance and monitoring water content using TDR

TDR serves effectively for monitoring hydrological water balance, measuring agricultural or forest water use efficiency, and monitoring changes in water content for irrigation scheduling. This type of monitoring requires rapid, reproducible recovery of data from a number of representative locations. These requirements led to the development of automated analyses of the TDR trace for water content (Zegelin et al., 1989; Baker and Allmaras, 1990; Heimovaara and Bouten, 1990; Herkelrath et al., 1991; Skaling, 1992 and Zegelin et al., 1992). Associated with this automation was the development of multiplexing, or switching capabilities that allow periodic measurement of TDR probes at multiple locations by a single TDR instrument. A number of custom systems have been developed and used in recent years (Zegelin et al., 1989; Baker and Allmaras, 1990; Heimovaara and Bouten, 1990; Herkelrath et al., 1991; Cassel

et al., 1994). Most commercial TDR-based instruments now offer automated water content analysis as a part of the basic instrument with the multiplexing capability as an option.

TDR for monitoring water content must be approached in relation to three different scales of measurement: (a) Point or near point (i.e. regions within 1 m and/or within a soil profile) with replication and real-time continuous data recovery; (b) Spatial replication at tens of meters (up to 100 m) with real-time continuous data recovery; (c) Spatial replication >100 m. Options (a) and (b) are currently available using an appropriate TDR instrument, multiplexers and probes, instruments measuring over separations greater than 25 m between instrument and probe require better quality cables and give rise to associated cost increases. At the present time option (c) would require separate instruments for widely spaced locations or the loss of real-time possibilities for data retrieval. Intermittent sampling would be dependent on operator availability (Topp et al., 1996a).

Field water balance studies using TDR have been carried out at differing scales, referenced to rain gauges, weighing lysimeters and Bowen ratio mass balances (Herkelrath et al., 1991; Van Wesenbeeck and Kachanoski, 1988 and Zegelin et al., 1992). Over a one-year period, in a forested plot, Herkelrath et al. (1991) measured the water budget using vertically installed 50 cm long TDR probes. They found excellent concurrence for the timing of rainfall events with only a 13% discrepancy between TDR and precipitation gauge. Using TDR (0–0.8 m deep probes) and weighing lysimeter (0–2.0 m deep) in a wheat field, Zegelin et al. (1992) compared the changes in stored water for two periods, one of 6 days involving wetting and drying cycles and the other of 16 days involving depletion by evapotranspiration. The amounts of wetting and drying were similar for both techniques with an average deviation of less than 10% in both periods. Changes in TDR-measured soil water storage below the turfgrass cover showed excellent temporal coincidence and response times with those measured by weighing lysimeters (Young et al., 1997). During daily irrigation of the turfgrass, the TDR probes underestimated the amount of water added and lost, but this was probably due to the daily storage and loss of water from within the surface thatch of the turfgrass

for which the TDR could not account. By contrast, during an unirrigated 6 days interval, the TDR probes were able to measure the water loss, with the longest probes recording 96% of the lysimeter-measured loss. Water exchange between soil and atmosphere was measured by TDR and verified intermittently by weighing microlysimeters, and by Bowen ratio and rain gauges (Baker and Spaans, 1994). For bare soil, the 21-day water balance for these systems was in agreement to within 10% when drainage beyond the TDR probe depth was taken into account.

In addition to total mass transfers, TDR has been used to indicate the distribution of water extraction with depth. Using vertical probes of differing length, Young et al. (1997) characterized the increasing amounts of water being taken up from deeper layers as the soil dried during the 6 days drying interval. Zegelin et al. (1992) compared hourly water content changes measured using TDR, lysimeter, and Bowen ratio techniques over a 12 h period. The TDR-measured soil water losses agreed very well with those of the other two techniques until 11:00 h, after which the results diverge. A feasible interpretation is that the wheat extracted water from depths above 0.8 m while the evaporative demand was low, i.e. before 11:00 h, and used deeper roots to extract water after this time, which was not detected by the 0.8 m length TDR probes.

Although plant roots have been recently identified as responsible for over half of the water transfer and much of the chemical relocation within the soil profile, their effects are often neglected in water and solute transport studies (Clothier and Green, 1997). Clothier and Green (1997) have shown examples of how TDR is one several emerging technologies being used to overcome the neglect of roots as agents of mass energy transfer in soil (Wraith and Baker, 1991; Topp et al., 1996a; Green et al., 1997).

Lateral spatial variability of water content is a common occurrence that may arise from soil factors, such as, topography, texture, structure, etc. as well as from crop factors. Attempts to characterize such variability are relatively recent. For example, Van Wesenbeeck and Kachanoski (1988) installed 100 TDR probes to a depth of 20 cm in an equally-spaced linear array across 25 maize rows and over a depth of 0–20 cm. In addition to showing the distributing effects of leaf and stem flow on rainfall infiltration

Table 1
Mean soil water recharge from an 8.2 mm rainfall on day 220
(from Van Wesenbeeck and Kachanoski, 1988)

Sampling location	Soil water storage		
	1 h before rainfall, S_i (mm)	0.5 h after rainfall, S_f (mm)	Recharge, $S_f - S_i$ (mm)
Row	54.2	64.4	10.2a ^a
Quarter row	52.4	60.3	7.9b
Inter-row	55.8	61.4	5.6c

^a Values in the same column with the same letters are not significantly different at the 0.05 probability level.

(Table 1) and subsequent water uptake by roots, they further calculated an average TDR-measured soil water recharge of 7.9 mm, which differs by only 5% from the 8.2 mm recharge measured by a rain gauge.

After cutting grass hay in late summer, we made TDR measurements with repeated insertions to a depth of 0.15 m using a portable probe similar to that described by Topp et al. (1984). From 750 locations on a $10 \times 10 \text{ m}^2$ grid, a mean water content of $0.236 \pm 0.024 \text{ m}^3 \text{ m}^{-3}$ and ranging from 0.16 to 0.32 was recorded (Topp et al., 1996b). In spite of uniform cropping practices and level topography (<0.5 m elevation difference over 7.44 ha), a distinct spatial pattern emerged (Fig. 2). As we have found less variability during wetter spring conditions, this spatial variation at the end of a growing season includes effects of spatially variable losses (e.g. water use and drainage).

With TDR it is now possible to determine hydrological balances more effectively and efficiently, and obtain spatial and temporal details of mass transfers that were not feasible a decade ago.

4. Monitoring solute mass transfer with TDR

Since both water content and bulk electrical conductivity (EC) of the soil are determined on the same measured volume, bulk EC is readily converted to pore water EC using the measured water content. If pore water EC is due to a single ionic (electrolytic) solute then pore water EC can be converted to solute concentration.

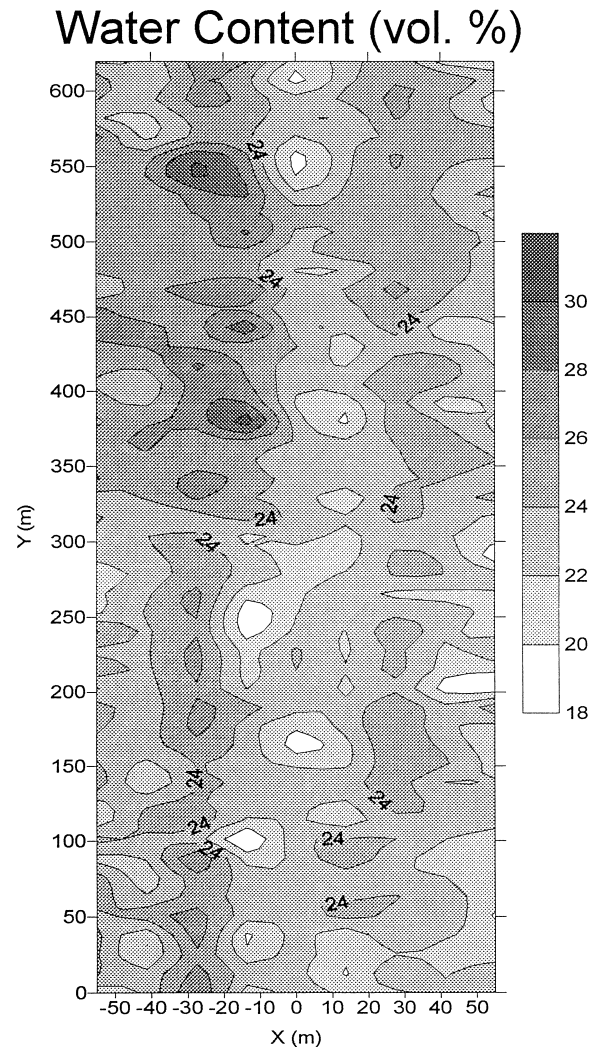


Fig. 2. The spatial pattern of soil water content in the 0–15 cm depth range of a 7.44 ha level, clay loam field after cutting hay.

Using vertical TDR rods, Kachanoski et al. (1992) measured one-dimensional transport of an ionic tracer solute under steady-rate, saturated flow in a laboratory column and in the field under steady-rate, unsaturated flow. A constant EC was measured until the dispersing tracer pulse reached the end of the rods, demonstrating the ability of TDR to measure average pore water EC in spatially and temporally uniform water content conditions. This measurement is possible even if the solute concentration is varying spatially. Further, in both field and laboratory studies, TDR gave the mass

flux of the tracer past the ends of the rods. At a heterogeneous field site, using vertically-installed TDR rods, Rudolph et al. (1996) found that mass flux measured with TDR was in good agreement with that obtained from solution samplers. TDR appears to measure solute transport, even when water content and solute concentrations vary spatially, but not necessarily the average pore water EC.

In layered soil, horizontally placed rods provided profiles of solute mass along a soil column during solute transport (Ward et al., 1994). Over a wide range of tracer concentrations, at four spatially uniform water contents, Ward et al. (1994) found a linear relationship between TDR-measured bulk EC and the concentration of solute. On soil columns ranging from very fine sand to clay loam, Risler et al. (1996) imposed cycles of wetting and drainage. From this, they found a linear relationship between TDR-measured bulk EC and the product of water content by pore water EC. These TDR-measured water content dependence relationships will need more confirmation, as they are not in agreement with the model of Rhoades et al. (1976), where the water content has a quadratic influence on bulk EC.

Although the application of TDR for simultaneous measurement of both water content and electrical conductivity has improved substantially our ability to monitor solute transfer, further work is required before TDR measurements can be made reliably under the conditions of transient water content and/or solute concentration.

5. Measuring oxygen concentration in soil

The transport of oxygen from the atmosphere into the soil is essential to root respiration, microbial activity and all oxidation processes. Given that water and air share the same pore space in the soil, the amount and movement of water exerts a strong influence on the amount and movement of air. Thus the ability to determine accurately the oxygen concentration and fluxes in the root zone depends critically on the ability to measure water content. Using non-destructive and repeatable measurement of water content by TDR in combination with in situ oxygen cathode measurements (Willey, 1974) we have been able to monitor the root zone oxygen concentration

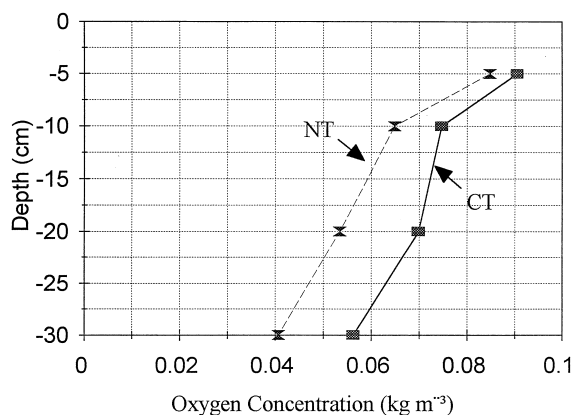


Fig. 3. Oxygen concentration (kg m^{-3} total soil volume) in a clay loam soil beneath maize rows 9 weeks after planting into no-till (NT) and conventional moldboard plow (CT) tillage.

during the growing season. Beneath maize rows at 9 weeks after planting, we found that the upper 0.3 m of a clay loam soil had up to 25% less oxygen under no till management (Fig. 3) compared to that under conventional moldboard plowing. Periodic measurements showed that these differences persisted for long periods, even in a relatively dry growing season.

6. Summary and conclusions

The development of TDR has greatly enhanced our ability to measure and monitor the storage and movement of water, solute and air in the crop root zone. Evaluations of TDR for hydrological mass balances have shown TDR comparable to weighing lysimeters, Bowen ratio and rain gauges but with an improved response time and greater operational flexibility. For solute transport the TDR is proving to be very effective and causes less disruption of the flux pattern than solution samplers. The application of TDR to transient solute flux measurements will depend on more rapid response in data collection and improved probe design. The repeatable, in situ, non-destructive measurements allowed by TDR have opened greater possibilities to augment data for other transfer processes, such as oxygen diffusion.

The importance of TDR in mass and energy transfer has only begun to be realized. The use of TDR should increase significantly as continuing developments in

instrumentation both reduce the instrument costs and increase the flexibility and applicability of the TDR technique. In addition, high frequency electromagnetic instruments, based on principles learned from TDR applications, are beginning to appear on the market. Such approaches offer less flexibility but have the advantage of lower cost. However, their validity is not yet clearly established. Continued development of TDR and related “spin off” techniques such as high frequency methods will undoubtedly contribute substantially to our understanding and exploitation of mass and energy transfer in soil.

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